12-4 Application of Programming [1] (Python)

Point!

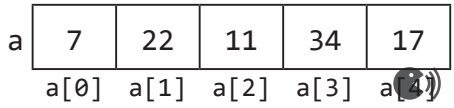
(List)

(1) ([1]List) : A collection of data items arranged in sequence. Data can be managed collectively.

List declaration	List name = [data0, data1, data2, …]
Adding elements to the end of an list	List name.append(data)

<Example> Character string list `week = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']`
List of numbers `number = [7, 22, 11, 34, 17]`

(2) ([2]Element) : Each value included in an list. Displays the list name and its place number
Using ([3]Index)(element numbers) makes it possible to retrieve elements from a list. Note that Index do not start from 1, but from ([4]0).



(3) List operation : Allows list element to be managed using a single index.
<Example> List declaration and method for accessing elements
Using a variable as a index such as in `a[i]` makes it possible to specify elements with the variable. This is convenient when processing elements one at a time using a "for" statement.

01	`a = [7, 22, 11, 34, 17]`	Declares list "a" and assigns a numerical value to it.
02	`for␣i␣in␣range(0,5,1):`	Repeats while increasing variable `i` from `0` to ([5]4) in increments of `1`.
03	`␣␣␣␣print(a[i])`	Displays elements from list "a" that satisfy the condition ([6]`a[i]`).

Execution result
```
7
22
11
34
17
```

<Example> Adding an element to the end of a one-dimensional list

01	`a = [ ]`	Declares the list "a".
02	`a.append(1)`	Adds ([7]1) to the end of list "a".
03	`a.append(4)`	Adds ([8]4) to the end of list "a".
04	`a.append(9)`	Adds ([9]9) to the end of the list "a".
05	`print(a)`	Displays the values for list "a".

Execution result
```
[1, 4, 9]
```

(4) ([10]Two-dimensional list) : An list that manages data using indexs in both the row and column directions. An element at row "i" and column "j" would be specified using the two indexs as `a[i][j]`.
<Example> Definition of a two-dimensional list and the method for accessing elements

01	`a=[['A', 'B', 'C'], ['D', 'E', 'F'],]`	Declares list "a" and assigns a character string to it.
02	`print(a[0][0])`	Displays the value for ([11]`a[0][0]`).
03	`print(a[1][2])`	Displays the value for ([12]`a[1][2]`).

Execution result
```
A
F
```

Warm Up

Answer the following questions.

(1) For programs **A** and **B** below, give the values displayed when each is executed.

A
```
a = [57, 16, 29, 44]
print(a[2])
```

B
```
a = [[1, 2, 3],
     [4, 5, 6],
     [7, 8, 9]]
print(a[2][1])
```

(2) The following program is designed to find the minimum value among the elements in list "a". Fill in the blanks **A** and **B** with the appropriate characters or numbers to complete the program.

```
a = [34, 52, 11, 40, 17]
min = a[0]
for i in range(1, [ A ], 1):
    if a[i] < min:
        min = [ B ]
print(min)
```

(3) Fill in the blanks **A** to **E** in the following program with the appropriate characters or numbers to complete the program so that it displays the "Execution result" as shown.

```
a = [['A', 'B', 'C', 'D'],
     ['E', 'F', 'G', 'H']]
for i in range([ A ], [ B ], 1):
    for j in range([ C ], [ D ], 1):
        print([ E ])
```

Execution result

```
A
B
C
(Omitted)
H
```

Explanation

(1) Note that list index start at `0`, not `1`. In addition, an element in a two-dimensional list can be represented as `a[i][j]`, where "i" indicates the row and "j" indicates the column. Therefore, **A**: `29`, **B**: `8`

(2) Set "`min`" as the variable that stores the minimum value of the list "`a`" so it is `a[0]`. Next, examine each element of list "`a`" sequentially, and if `a[i]` is smaller than `min`, update `min` to that value.
Therefore, **A**: `5`, **B**: `a[i]`

(3) **A**: `0`, **B**: `2`, **C**: `0`, **D**: `4`, **E**: `a[i][j]`

Try

Answer the following questions.

(1) For programs **A** to **C** below, give the values displayed when each is executed.

A
```
a = [1, 4, 9, 16, 25]
print(a[3])
```

B
```
a = [ ]
a.append(8)
a.append(28)
print(a[1])
```

C
```
a = [['A', 'B', 'C', 'D', 'E'],
     ['F', 'G', 'H', 'I', 'J'],
     ['K', 'L', 'M', 'N', 'O']]
print(a[2][1])
```

(2) The following program finds the total of the elements in list "a". Fill in the blanks **A** to **C** with the appropriate characters or numbers to complete the program.

```
a = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
sum = 0
for i in range( A , B , 1):
    sum = sum + C
print(sum)
```

(3) The following program finds the maximum value among the elements in list "a". Fill in the blanks **A** to **C** with the appropriate characters or numbers to complete the program.

```
a = [7, 22, 11, 34, 17]
max = 0
for i in range(0, A , 1):
    if a[i] B max:
        max = C
print(max)
```

(4) Fill in the blanks **A** to **E** in the following program with the appropriate characters or numbers to complete the program so that it displays the "Execution result" as shown.

```
a = [['A', 'B', 'C', 'D', 'E'],
     ['F', 'G', 'H', 'I', 'J']]
for i in range( A , B , 1):
    for j in range( C , D , 1):
        print( E )
```

Execution result

```
A
B
C
(Omitted)
J
```

Exercise

1 Answer the following questions.

(1) For programs **A** to **C** below, give the values displayed when each is executed.

A
```
a = [0, 1, 2, 3]
print(a[1])
```

B
```
a = ['A']
a.append('B')
a.append('C')
print(a[1])
```

C
```
a = [['Japanese', 'Mathematics', 'English'],
     ['Social studies', 'Science', 'Information studies']]
print(a[1][2])
```

(2) The following program is designed to find the minimum value among the elements in list "a". Fill in the blanks **A** to **C** with the appropriate characters or numbers to complete the program.

```
a = [24, 43, 9, 25, 17]
min = [ A ]
for i in range(0, [ B ], 1):
    if a[i] < min:
        min = [ C ]
  print(min)
```

(3) The following program counts the number of elements in list "a" that are greater than 10. Fill in the blanks **A** to **C** with the appropriate characters or numbers to complete the program.

```
a = [12, 8, 9, 13, 11]
count = 0
for i in range([ A ], [ B ], 1):
    if [ C ] > 10:
        count = count + 1
print(count)
```

(4) Fill in the blanks **A** to **E** in the following program with the appropriate characters or numbers to complete the program so that it displays the "Execution result" as shown.

```
a = [['Japanese', 'Mathematics', 'English'],
     ['Social studies', 'Science', 'Information studies']]
for i in range([ A ], [ B ], 1):
    for j in range([ C ], [ D ], 1):
        print([ E ])
```

Execution result

```
Japanese
Mathematics
English
Social studies
Science
Information studies
```